DECISION / REQUESTED ZERO MOUNTING ANGLE

Angled branches, mirrored inversion

Pulley ratios and nearly equal links searched from the screenshot; alpha fixed at zero.

WINDOWS EVALUATED

26.46 million

Two distinct window families

PRACTICAL BRANCH

12.02 degrees

350/350 mm; 7:4 guide

MAXIMUM BOW

9.66 mm

2.91% of its endpoint chord

The strict requested combination is not attained: with alpha=0 and the original assembly phase phi0=0, a normal-to-inverted path has a vertical tangent at the fold. A symmetric full stroke has a vertical chord. Ratio and link-length changes can produce a useful angled suspension branch, but its inverted branch tilts the opposite way.

Requirement / interpretation	Finding
10-15 deg normal working path	Many candidates; straightness and tangent spread vary
Same tilted line through full inversion	Impossible under fixed alpha=phi0=0 ideal equations
Near-equal links + coplanar 200 mm wheel	Every searched full inversion fails projected hip clearance
Pulley indexing allowed as a relaxation	phi0=-25 deg, guide 2 gives a true 12.5 deg line; fold collision remains

Practical example: 325 mm total vertical capacity with 20 mm at each end gives 285 mm usable. Its 9.66 mm bow is acceptable only if that approximately 3% deviation suits the design. It fails the stricter illustrative 2% preference. The straightest tested branch uses longer links near the 450 mm search boundary.

This report preserves the earlier study and Motion Lab tooling. Fully invertible here means continuous ideal guide motion from below to above the hip; it is not a demonstrated ground flip or physical wheel/chassis clearance. Dual-rope remains IMPOSSIBLE / rejected.

The fold tangent stays vertical

Jonathan variables retained: q=theta2, guide r=ra/rb, link lengths a=L2 and b=L3.

With t=q-pi and k=r-1, alpha=phi0=0:

```
x = a cos(t) - b cos(k t)
z = a sin(t) + b sin(k t)
x(-t) = x(t); z(-t) = -z(t)
C(pi) = (a-b,0)
C'(pi) = (0,a+k b)
```

The horizontal coordinate is even and height is odd. Upper/lower inversion branches mirror across the horizontal hip plane; their local tilts have opposite signs. At the fold the tangent is vertical, with a finite positive Jacobian for the searched ratios. A chord tilted 10-15 degrees over an asymmetric interval does not remove that vertical tangent.

Exact constraint	Consequence
r=2, a=b, phase 0	The only exact nondegenerate straight path is vertical
r not 2	Distinct angular frequencies produce curvature
r=1+sqrt(a/b)	Cancels central quadratic lateral curvature; centre remains vertical
beta=13.5 deg	Defines contact/load orientation; it does not rotate the wheel-centre path

Centre series:

```
x = (a-b) + (b k^2-a)t^2/2 + O(t^4)
z = (a+b k)t - (a+b k^3)t^3/6 + O(t^5)
```

Changing ratio can make the centre locally straighter, but cannot create a nonzero central tilt.

Proof scope: constant positive links and ideal fixed-ratio guide. Variable-ratio transmissions or an additional linkage are different architectures and were not silently substituted.

SEARCH / WHAT WAS ENUMERATED

A broad reproducible sweep

Search counts describe geometry-plus-window combinations, not hardware designs or global proof.

Variable	Domain / policy
Lower-link L3	200-450 mm in 10 mm steps, plus 275/325/375/425
Link ratio L2/L3	0.85-1.15 in 0.005 steps; equal/5%/10%/15% tiers
Guide ratio ra/rb	1.4-2.6 in 0.0025 steps, plus exact 5/3; rb 60 mm
Mounting / assembly phase	alpha 0, phi0 0 in the strict search
Normal start / total	theta2=100/105/110/115/120; total 300/325/350 mm
Crossing-window allocation	Above-hip fractions 20/35/50/65/80%; exact heights
Pose verification	33 coarse poses; finalists 2001 dense poses; TLS uniform height
Screenshot values retained	rw 200; drive radii 35; beta 13.5; Fy=Fz100; m1=10, others 0

Count	Observed result
Normal-window combinations	13,230,900
Crossing-window combinations	13,230,900
Normal 10-15 deg + branch-clearance passes	870,568
Coarse normal <=2% bow, <=15 deg tangent span	8,080
Crossing-window candidates meeting both limits	0
Full coplanar inversion clearance passes	0, also excluded by the exact fold-distance argument

Full height-monotone inversion is checked across theta2=100-260 degrees. Normal branch candidates retain positive vertical Jacobian and projected wheel/hip margin >=20 mm. The 20 mm geometric reserves are not spring compression. Ranking minimizes maximum orthogonal chord error, then tangent spread, then link sum. These are bounded grid winners.

Floating grid coordinates are rounded before deduplication. Quality counts use 33 coarse poses. Source, domains, metrics and selected curves are retained in inversion-path-results.json; raw runs stay ignored.

NORMAL SUSPENSION / EQUAL AND NEAR-EQUAL

What the branch search found

All rows below have alpha=phi0=0. Tilt is the endpoint chord, not every local tangent.

L2/L3 mm	Guide	Total mm	Chord deg	Bow mm	Tangent span deg	Wheel/hip margin mm
450/450	1.8000	300	10.00	5.22	8.16	224.8
450/450	1.7925	325	10.07	6.16	8.91	197.3
450/450	1.7875	350	10.01	7.11	9.58	170.6
459/450	1.8050	325	10.00	6.10	8.83	208.7
486/450	1.8375	325	10.01	6.06	8.81	242.0
517.5/450	1.8750	325	10.00	6.06	8.84	282.0
450/450	1.7400	325	12.45	7.70	10.98	180.4
472.5/450	1.7675	325	12.44	7.56	10.81	208.2
450/450	1.6825	325	14.96	9.43	13.23	160.7
322.5/375	1.7375	325	10.03	7.73	11.60	79.7
300/300	1.8175	325	10.02	9.83	16.30	28.1
350/350	1.7675	325	10.08	8.04	11.73	30.3
400/400	1.7825	325	10.05	6.94	10.08	114.1

The straightest branches sit near the maximum link-length bound and the lower 10-degree angle target. Increasing size narrows the joint-angle window for the same 325 mm height, reducing bow. This does not establish a structural, mass or motor optimum. The 350/350 mm example preserves equal links and uses a simple 7:4 ratio at greater curvature.

Link closeness uses |L2/L3-1|. A 5% link mismatch improves the best 325 mm bow only slightly in this domain. Full inversion clearance fails for every row, even when its normal working branch passes the projected filter.

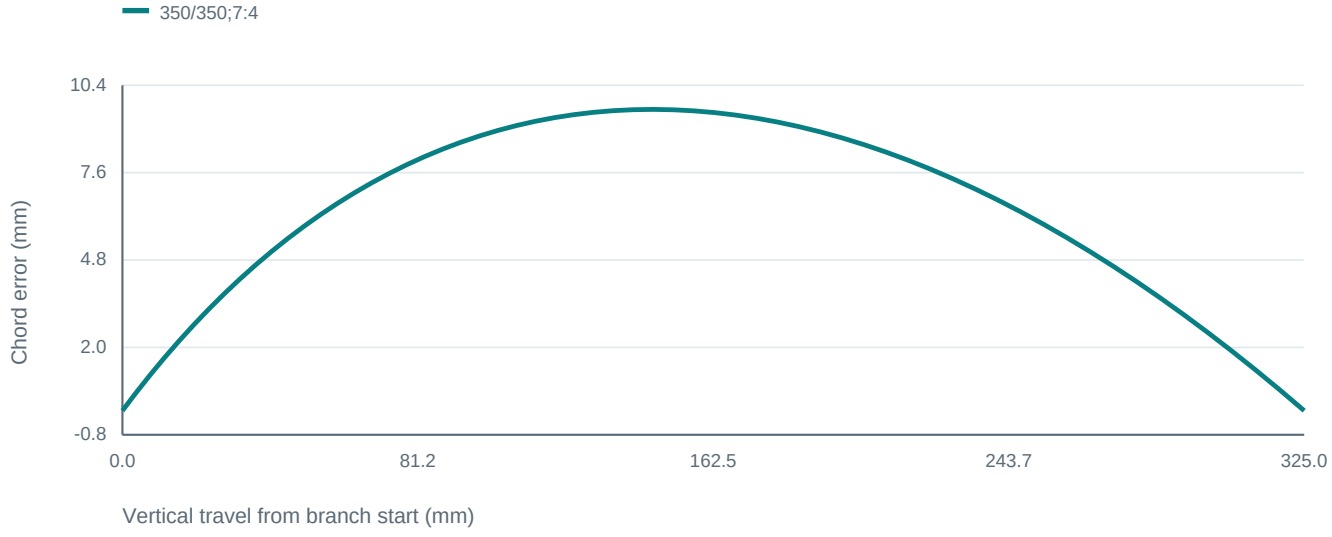
PRACTICAL COMPARISON / SIMPLE GUIDE RATIO

Equal links with a 7:4 guide

Illustrative compromise: 350 mm links, zero mounting angle, zero assembly phase.

Parameter / requirement	Recorded value
L2/L3; ra/rb	350/350 mm; 105/60 mm -> 7:4
Physical theta2 interval	115 to 155.512438 degrees
Total / usable height	325 / 285 mm; 20 mm reserved at either end
Endpoint chord / TLS angle	12.0201 /11.7500 degrees
Maximum orthogonal chord bow	9.6587 mm (2.907%)
Local tangent range	6.2697 to 20.4741 degrees
Minimum normal branch wheel/hip margin	55.7165 mm
Full 100-260 degree sweep	Normal and inverted heights, finite vertical Jacobian; fold radius 0

Bow relative to the endpoint chord

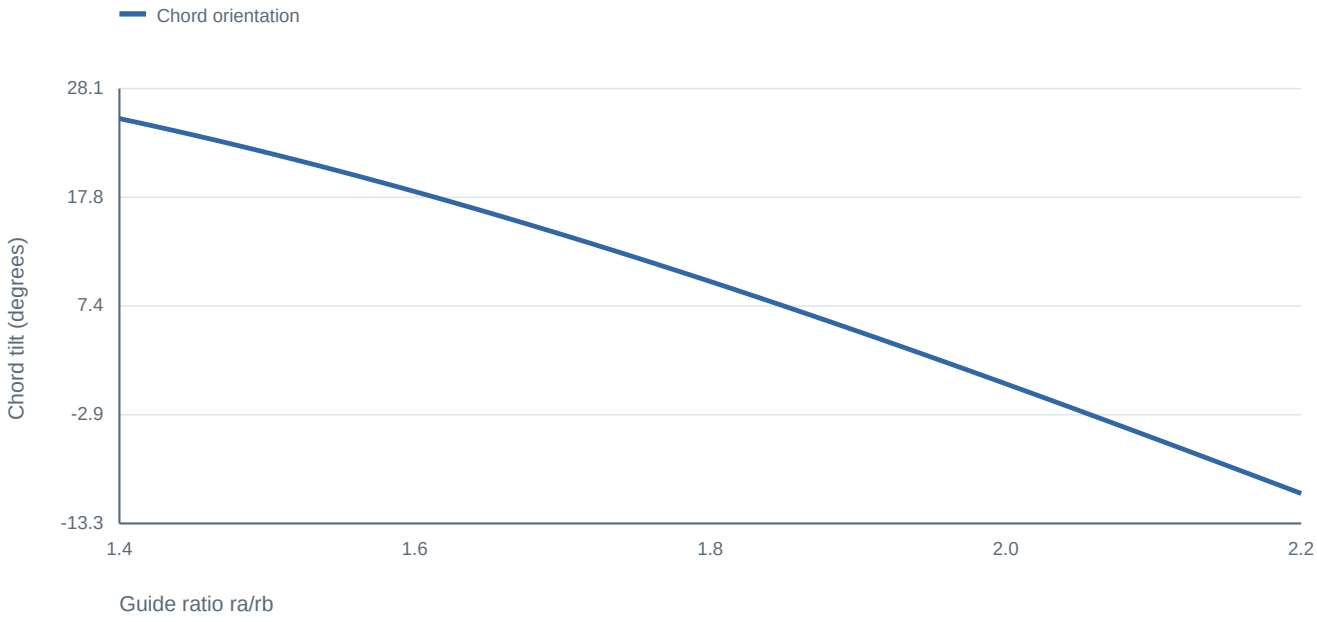


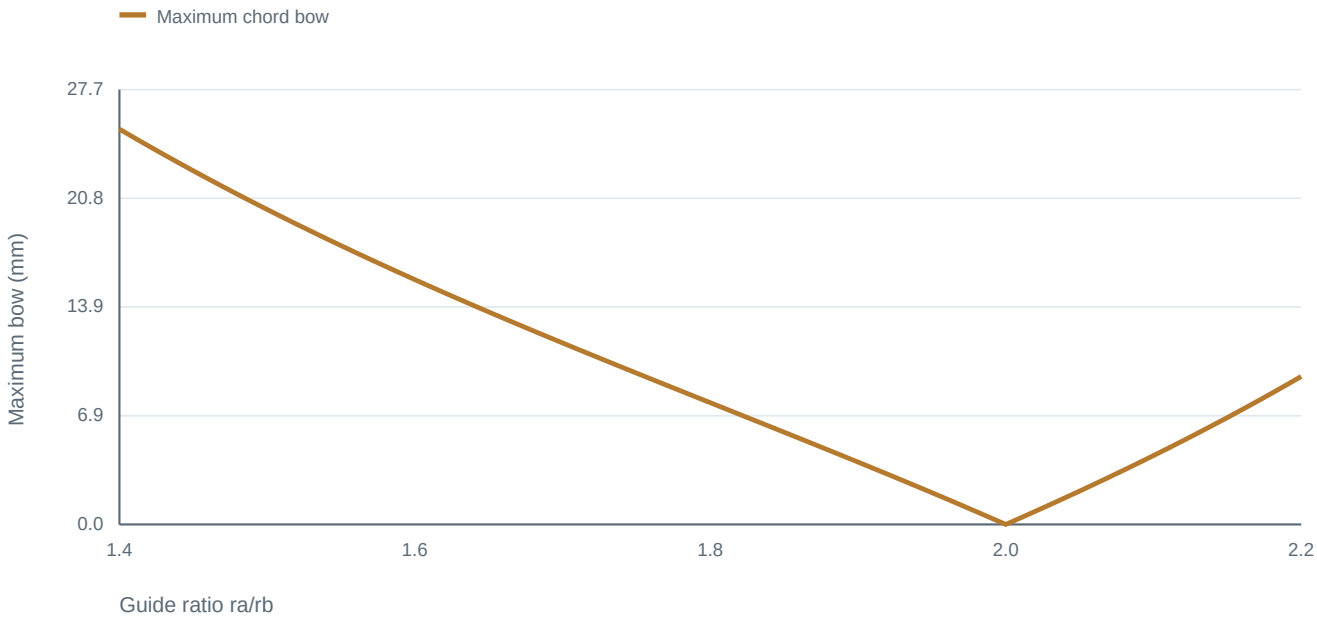
This candidate conditionally suits a design that accepts about 10 mm bow over 325 mm. It does not meet the illustrative 2% straightness preference, and the local tangent is not confined to 10-15 degrees. Full inversion is only an ideal linkage result until axial wheel/chassis routing is defined.

SENSITIVITY / RATIO TUNES ANGLE AND BOW

Ratio alone introduces curvature

Equal 350 mm links, start 115 degrees, exact 325 mm vertical stroke; alpha=phi0=0.



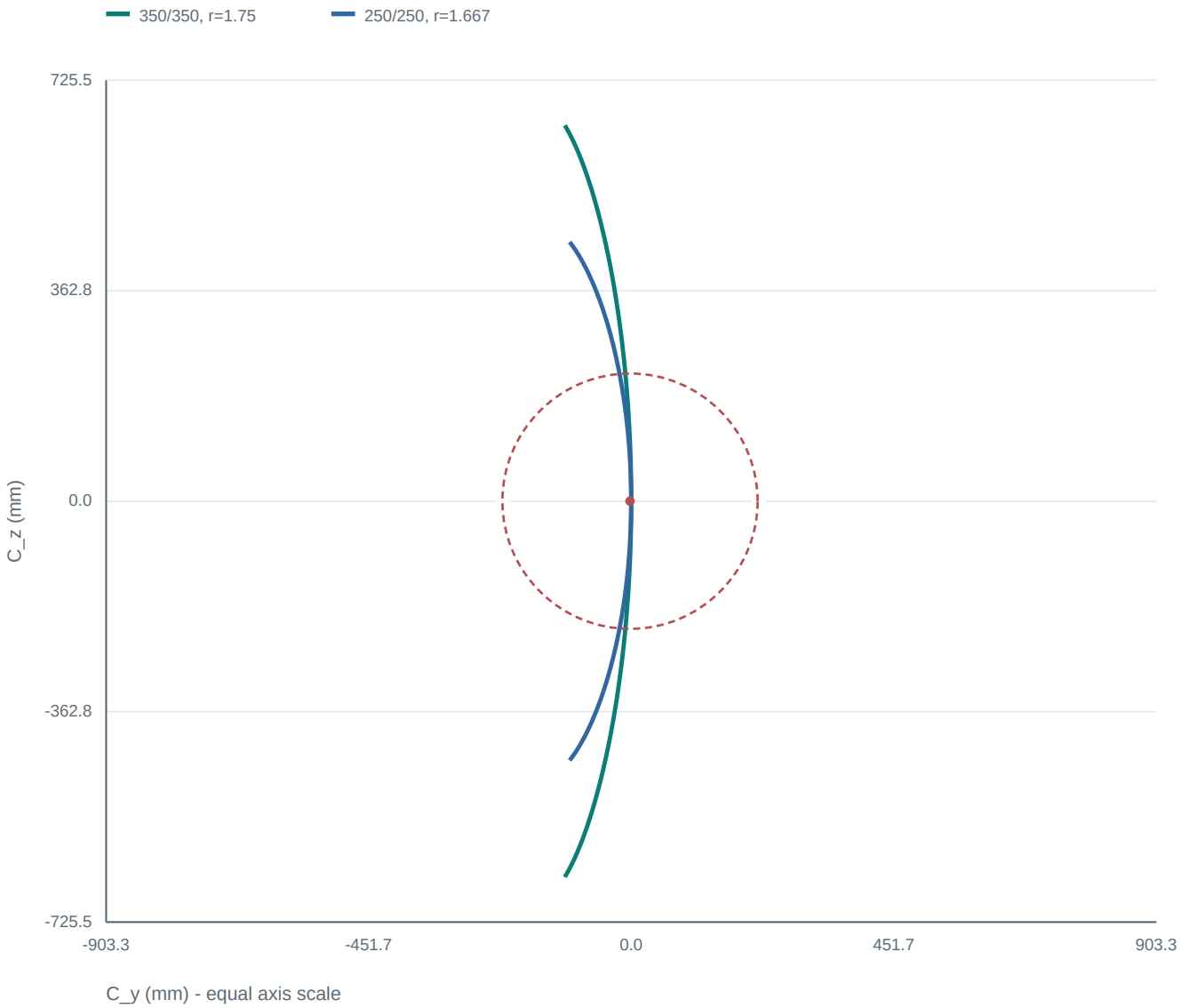


Moving away from 2:1 changes the wheel-centre path orientation over a selected branch and introduces bow. The exact equal-link 2:1, phase-zero case is vertical. Different qcd/qef wheel-drive ratios do not change the wheel-centre locus; only qab affects this guide geometry. Keep nominal wheel spin K distinct from path straightness.

FULL INVERSION / THE TWO BRANCHES

The inverted branch leans oppositely

World wheel-centre coordinates, equal axis scales; red 220 mm disk is projected exclusion.



The normal 350/350 branch leans 12.02 degrees by its endpoint chord. Its mirrored inverted suspension branch leans -12.02 degrees. The complete symmetric sweep has a vertical chord, with a vertical tangent at the folded position. A single angled line is not maintained through the inversion.

All phi0=0 full strokes pass C=(L2-L3,0). For equal links, wheel centre C coincides with hip A. With <=15% mismatch and L3<=450 mm, |L2-L3|<=67.5 mm, below the 220 mm projected exclusion. Axial separation could change physical clearance but is not specified.

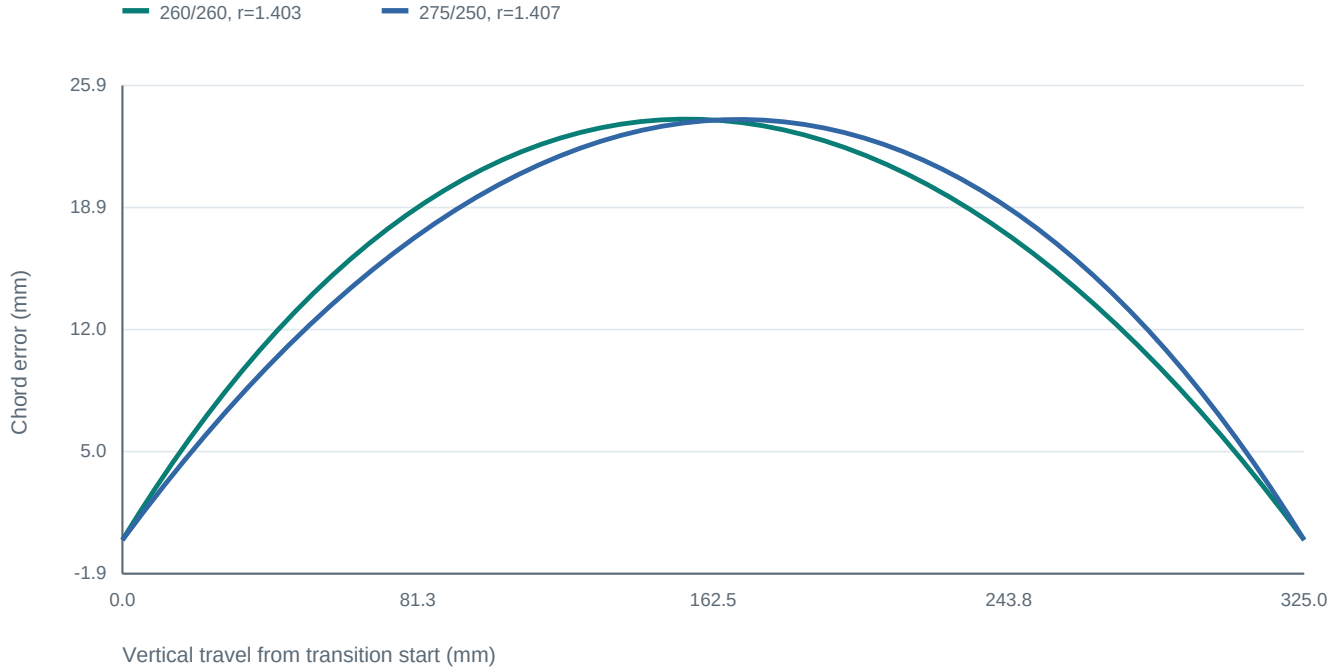
MISLEADING CHORD / FINITE CROSSING WINDOWS

A tilted chord can hide a large bow

300-350 mm total inversion transitions also searched, with 20-80% above-hip allocation.

L2/L3 mm	Guide	Chord deg	Bow mm	Tangent span	Hip distance
260/260	1.4025	10.003	23.942	33.605	0.0
275/250	1.4075	-10.002	23.922	33.523	25.0

Endpoint angle 10-15 deg; large internal deviation



The best sampled 325 mm crossing examples bow about 24 mm, or 7.25% of chord, and have roughly 33.5-degree tangent variation. None of the crossing candidates meets both <=2% bow and <=15-degree tangent spread. The midpoint vertical tangent prevents an everywhere 10-15-degree path, even if the endpoints appear promising.

A finite normal suspension window and a normal-to-inverted window are distinct tasks. The full inversion envelope of 350 mm links has approximately 1296 mm vertical extent; its selected suspension branch has 325 mm total. Do not call both325 mm travel.

EXPLICIT RELAXATION / KEEP ALPHA ZERO

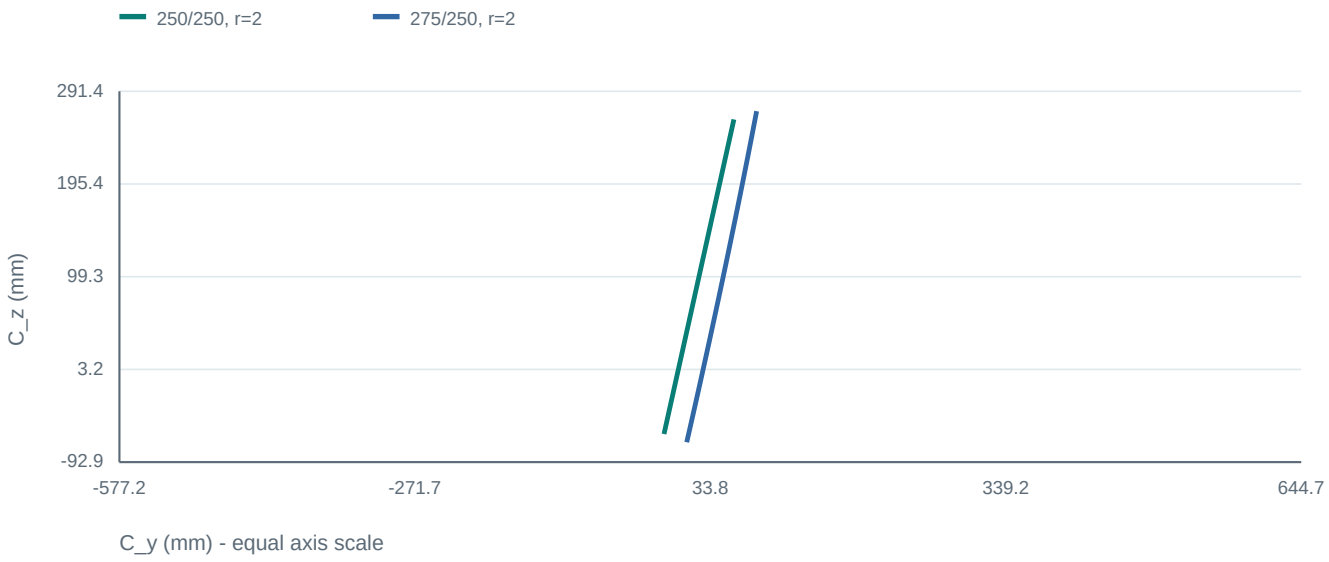
Pulley indexing can tilt a true line

Supplementary alternative: phi0 is allowed to change; it is not the strict phase-zero search.

For equal links L and guide 2:
`C = 2L sin(t-phi0/2) [-sin(phi0/2), cos(phi0/2)]`

Thus phi0=-20 to -30 degrees produces a true 10-15-degree line with alpha=0. This sets the lower-link assembly/belt phase. Mounting alpha instead rotates both links and their frame; the wheel-centre line can coincide while the knee path differs.

Case	Vertical mm	Chord deg	Bow mm	Tangent range
250/250,2:1,-25deg	325.998	12.500	0.000	12.5 to 12.5
275/250,2:1,-25deg	343.099	11.895	1.544	10.762 to 12.859

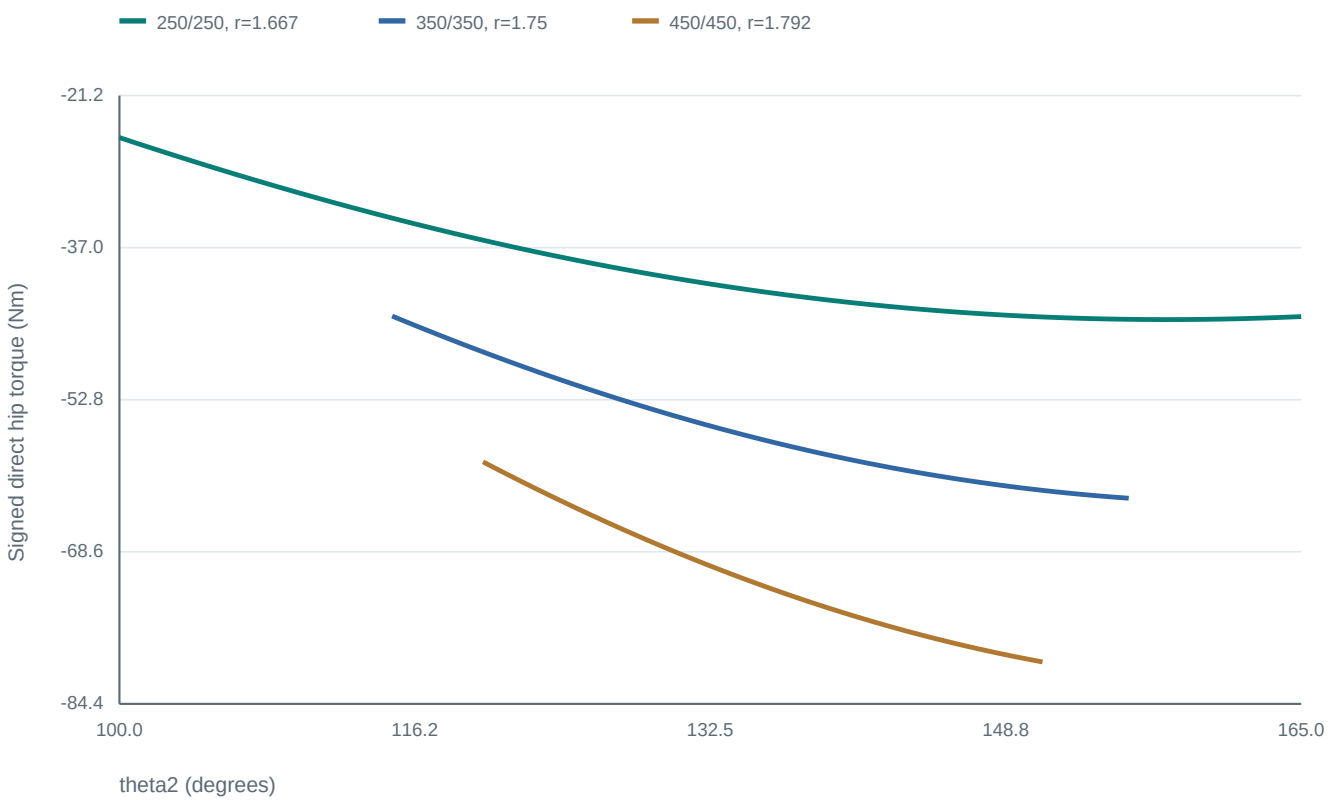


The phase alternative relaxes the request for the offset to arise only from ratio and length. It still folds to zero hip distance for equal links, or 25 mm for 275/250. It therefore does not solve coplanar 200 mm wheel clearance. It is a candidate only with explicit indexing and unresolved axial packaging.

DIAGNOSTIC / INHERITED LOAD CONVENTION

Joint torque is a separate requirement

Original prescribed Fy=Fz=100 N; m1 at fixed A, other masses zero; no fitted spring.



The displayed torque follows the same fixed-A virtual-work load case as the original screenshot. It is not a standing 10 kg-per-leg dynamics prediction or a motor rating. An above-hip contact force is only a prescribed algebraic load, not proof of ground contact.

Coordinate relation:

`d(relative knee)/dq = -qab`

`tau_knee = -tau_hip/qab`

A smaller torque in another coordinate accompanies greater angular speed. The guide ratio is separate from any motor reduction; ideal work is conserved.

This follow-up selects path geometry. Existing spring/motor and impact estimates remain in the earlier report and do not silently transfer to these longer or fully inverted legs.

VERIFICATION / ACTUAL CONSTRAINED STEPPING

Math and engine checks

Independent equations, dense rechecks and actual Pymunk motion; physical collision excluded.

Independent review recomputed all 24 displayed cases from the original V2 equations at 20,001 poses. Endpoint classifications agree; maximum chord-error difference from retained 2001-pose checks is 0.0000044 mm. Existing model parity covers 8,784 comparisons with maximum 3.98e-13 mm/Nm discrepancy. Five new numerical tests cover derivatives, parity/fold identities, phase lines and exact un-clipped search capacity.

Actual engine case	2ms error mm	1ms error mm	0.5ms error mm	Fine check
450/450, r=1.792	0.1596	0.0792	0.0395	PASS
459/450, r=1.805	0.1640	0.0814	0.0406	PASS
322.5/375, r=1.738	0.1119	0.0555	0.0277	PASS
350/350, r=1.75	0.1201	0.0596	0.0297	PASS
275/250, r=2; phase -25	0.1197	0.0594	0.0296	PASS

Engine fixture: positive recorded masses, hip fixed, zero gravity/contact, 2-second quintic angular sweep, 800 solver iterations. Only initial poses are assigned; Space.step and real motor/pivot/gear constraints generate motion. Wheel position is compared with independent V2 coordinates at the actual femur angle. Coarse 2 ms runs exceed the 0.1 mm path tolerance in all five cases; 1 and 0.5 ms pass and converge.

A first zero-buffer height-driven check hit endpoint floating-point bracketing; this independent angular fixture avoids that inverse-height dependency. No old solver or tolerance was weakened. Self-collision/ground contacts are absent, so these passes do not negate the rejected projected-clearance result.

CONTINUATION / SOURCES AND DECISIONS

Choose which constraint may change

The strict outcome and the practical alternatives stay separate from hardware acceptance.

If approximate mirrored branches are acceptable, begin with equal 350 mm links and a 7:4 guide, while reserving about 10 mm path bow. If straightness dominates size, compare the 450 mm-bound candidates. If one same-direction 10-15-degree line through inversion is essential, the phase-zero fixed-ratio architecture must change.

Allowing assembly phase offers an exact tilted line at alpha 0. Axial packaging must separately keep a 200 mm wheel clear of the hip/chassis during folding. Neither a promising plot nor a collision-disabled engine run establishes that clearance.

Reproduction from the standalone report folder:

```
python inversion_search.py
python -m unittest -v test_inversion_search
python scripts/verify_inversion_geometry.py
python scripts/build_inversion_report.py
node scripts/verify_inversion_report.cjs
```

Primary project sources: Jonathan leg-solver/v1-kinematics.html, v2-quasi-static.html and v2-model.md; this app model.js and generalized_leg.py; new inversion_search.py, retained results and engine checks; docs/inversion-path-study.md. Read the revision map for the source base 865b5f1 and later publication.

Unresolved: axial offsets, chassis shape, shaft/gear/belt wrap and tooth counts, load path and spring routing through inversion, measured masses, structural limits and ground-contact behavior. The earlier dual-rope design remains IMPOSSIBLE/rejected; this report neither reuses nor revives it. No canonical robot validation is claimed.